

Zero Can Have  
an Inverse

0-1

Definition: We say a set  $F$  together with  $+$  and  $\cdot$  is a field if

- $x + y = y + x$  ,  $xy = yx$
- $x + (y + z) = (x + y) + z$  ,  $x(yz) = (xy)z$
- $x(y + z) = xy + xz$
- there is  $a \in F$  s.t.  $x + a = x \quad \forall x \in F$
- for all  $x \in F$  there is  $(-x) \in F$  ,  $x + (-x) = a$
- there is  $m \in F$  s.t.  $xm = x \quad \forall x \in F$
- as many  $x \in F$  as possible have  $x^{-1} \in F$  s.t.  $x \cdot x^{-1} = m$

In the rational numbers  $\mathbb{Q}$  :

- 0 has the property that  $0+r = r \quad \forall r \in \mathbb{Q}$
- each  $r \in \mathbb{Q}$  has an additive inverse  $(-r)$ :  $r + (-r) = 0$
- 1 has the property that  $1 \cdot r = r \quad \forall r \in \mathbb{Q}$
- Every  $r \neq 0$  has a multiplicative inverse,  $\frac{1}{r}$ , s.t.  $r \cdot \frac{1}{r} = 1$

**Question** When can the additive identity have a multiplicative inverse?

Suppose that  $a$  has a multiplicative inverse.

$$\Rightarrow \exists a^{-1} \in F \text{ s.t. } a \cdot a^{-1} = m.$$

Take any  $x \in GF$   $\rightarrow$   $[x+a = x]$   $\xrightarrow{a^{-1}}$   $[a^{-1}x + m = a^{-1}x]$   
 $\downarrow +(-a^{-1}x)$   
 $[m=a]$

Answer When the additive and the multiplicative identity are the same :  $a = m$

What does this say about the field?

Take  $x \in F$  :  $x = mx$

$$= ax$$

$$= (a+a)x$$

$$= ax + ax$$

$$= mx + mx$$

$$= x + x \quad \xrightarrow{+(-x)} \quad a = x \quad \Rightarrow \quad F = \{0\}$$